

Technical Cybersecurity Report

Analysis Using Metasploit framework for Network Scanning

By

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REPORT OUTLINE

Executive Summary

Background and Objectives

Methodology

3.1 Tool Configuration

3.2 Execution Process

3.3 Monitoring and Analysis

Findings and Analysis

4.1 Indicators of Compromise (IOCs)

4.2 Behavioral Analysis

4.3 Risk and Impact Assessment

Recommendations

5.1 Immediate Remediation Actions

5.2 Long-Term Mitigation

Conclusion

Appendix: Additional Data

1. Executive Summary

Overview:

The analysis using Metasploit and Nmap was conducted to perform a network scan on my ip address 192.168.40.1 which was gotten using the ipconfig command in the command prompt terminal of my system, as part of a security assessment task. Nmap, initiated within the Metasploit framework and independently utilized the options `-sC`, `-sV`, `-vv`, `-sL`, `-sn`, and `-Pn` to identify open ports, services, versions, host status and obtaining detailed information on the target's network susceptibility and vulnerability.

Key Findings:

Verbose output detailed service scripts and version information, while additional options confirmed the host's online status. Outdated software versions were mapped as potential security risks.

2. Background and Objectives

Project Context:

The analysis is aimed at examining the security posture of my ip address (192.168.40.1) targeted key host to observe risked services and potential vulnerabilities. Metasploit and Nmap were selected for the task due to their profound scanning abilities and capability in the world of penetration testing and ethical hacking.

Objective of the Tool Use:

The specific objective was to use Nmap within the Metasploit framework and Nmap to conduct a technical scan on my ip (192.168.40.1) identifying service versions, open ports and potential risk vectors for further implementations of security measures .

3. Methodology

3.1 Tool Configuration

Metasploit and Nmap Configuration:

- Open root terminal : `sudo su`
- Launched Metasploit: `msfconsole`.
- Database configuration:
`systemctl start postgresql,`

msfdb init,
msfconsole
db_status

- Configured Nmap within Metasploit using the db_nmap command.
- Target IP: 192.168.40.1.

Options:

- -sC: Script scan.
- -sV: Service/version detection.
- -vv: Very verbose output.
- -sL: List scan.
- -sn: Ping scan (no port scan).
- -Pn: No ping, treat host as online.

Simulated environment: Kali Linux 2024.1 with Metasploit Framework v6.3.

3.2 Execution Process

Initial Scan with -sC, -sV, -vv:

Commands:

- db_nmap -sC -sV -vv 192.168.40.1.
- db_nmap -sV 192.168.40.1.
- db_nmap -vv 192.168.40.1.
- The scan ran Nmap scripts, detected service versions, and provided detailed output, identifying active services on the target.

Comprehensive Scan with All Options:

- Command: db_nmap -sC -sV -vv -sL -sn -Pn 192.168.40.1.
- This scan combined all specified options, listing the target, skipping port scans where applicable, and providing verbose service details while assuming the host was online.

3.3 Monitoring and Analysis

System Monitoring:

Command db_nmap -sC 192.168.40.1.

Nmap within the metasploit framework captured open ports

- 135/tcp → MSRPC (Microsoft Remote Procedure Call)
- 139/tcp → NetBIOS-SSN (Used for file and printer sharing)
- 445/tcp → Microsoft-DS (SMB file sharing)
- 903/tcp → ISS Console Manager
- 5357/tcp → WSDAPI (Web Services for Devices)
- 8000/tcp → HTTP-Alt (Web server detected but lacks a title)
- 8089/tcp → Unknown (Possibly a custom service)

- 16992/tcp → AMT-SOAP-HTTP (Intel Active Management Technology)

Host Status: The target is up with a latency of 0.0044s. Which is good, 992 ports are filtered, meaning they didn't respond.

Network Monitoring:

Port 8089 (tcp), was monitored and detected to have Splunk SSL certificate which is valid until 2028. This suggests Splunk is running on the machine, possibly logging network activity.

Command db_nmap sc192.168.40.1

The scan detected a total of 8 open ports

- 135 MSRPC (Microsoft Remote Procedure Call)
- 139 NetBIOS-SSN (Windows File Sharing)
- 445 Microsoft-DS (SMB - Server Message Block)
- 903 ISS Console Manager
- 5357 WSDAPI (Web Services for Devices)
- 8000 HTTP-Alt (Web server detected)
- 8089 Unknown (Possibly Splunk logging service)
- 16992 AMT-SOAP-HTTP (Intel Active Management Technology)

Network Monitoring:

Host status:

latency (0.0049s). 992 ports are filtered, meaning they didn't respond often due to a firewall or security controls.

Command db_nmap -sV 192.168.40.1

The scan detected a total of 8 port open and running

- **135/tcp** – Microsoft Windows RPC, used for remote procedure calls between networked systems.
- **139/tcp** – NetBIOS-SSN
- **445/tcp** – Microsoft-DS,
- **903/tcp** – VMware Authentication Daemon,
- **5357/tcp** – Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP),
- **8000/tcp** – Splunkd HTTP service,
- **8089/tcp** – Splunkd SSL-enabled HTTP service,
- **16992/tcp** – Intel Active Management Technology (AMT) User Notification Service,

Host status: low latency (0.0019s). 992 filtered ports

4. Findings and Analysis

4.1 Indicators of Compromise (IOCs):

Detected IOCs:

IP Address: 192.168.40.1 (target host).

Open port:

- 135/tcp – Microsoft Windows RPC, used for remote procedure calls between networked systems.
- 139/tcp – NetBIOS-SSN
- 445/tcp – Microsoft-DS,
- 903/tcp – VMware Authentication Daemon,
- 5357/tcp – Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP),
- 8000/tcp – Splunkd HTTP service,
- 8089/tcp – Splunkd SSL-enabled HTTP service,
- 16992/tcp – Intel Active Management Technology (AMT) User Notification Service,

Service versions

Microsoft Windows RPC (Port 135) - No specific version.

Microsoft Windows netbios-ssn (Port 139) - No specific version.

microsoft-ds (Port 445) - Version not specified (uncertain detection).

VMware Authentication Daemon 1.10 (Port 903).

Microsoft HTTPAPI httpd 2.0 (Port 5357).

Splunkd httpd (Ports 8000 and 8089) - No specific version.

Intel Active Management Technology User Notification Service httpd 15.0.50.2633 (Port 16992).

4.2 Behavioral Analysis:

Service Actions:

Port 8000 has an active web service but lacks a proper title.

The requested resource points to /account/login, suggesting authentication is required.

Message signing is enabled and required → This helps prevent man-in-the-middle attacks in SMB communication.

4.3 Risk and Impact Assessment:

Potential Impact:

- SMB-related ports (445, 139) → are common targets for exploits—consider restricting access if not needed.
- Intel AMT (16992) → This port suggests remote management; if exposed improperly, it could allow full control over the system.
- Splunk services (8000, 8089) → If the logging system isn't securely configured, sensitive monitoring data may be accessible remotely or susceptible to brute-force or authentication bypass attempts.
- VMware Auth Daemon (903) → If exploited, it could lead to unauthorized remote authentication via VNC.

5. Recommendations

- Enumerate SMB vulnerabilities (445, 139).
- Assess Web Services (8000, 8089) : Investigate ports 8000 & 8089 using Burp Suite to test login security.
- Check Remote Management (16992) : If Intel AMT is enabled, review configurations to prevent unauthorized remote access.

5.2 Long-Term Mitigation:

- weekly scan should be encourage to monitor for new open ports and services
- Firewall should be set up for a stronger security
- Harden Authentication & Firewall Rules
 - Restrict and limit public access to exposed services via firewall or VPN policies.
 - Disable unnecessary and unused services to reduce attack surface.

6. Conclusion

Summary of Findings:

A total of 1000 ports were scanned in the analysis using Metasploit to scan my ip 192.168.40.1. 8 ports were detected as opened and actively running and the remaining 992 were flagged as filtered due to a firewall or security controls.

While no immediate threats were discovered, open ports such (445, 139), Intel AMT (16992) VMware Auth Daemon (903) if exploited could lead to remote access and lateral movement. therefore regular scan and monitoring system should be employ to actively note any change from this result.

Appendix: Additional Data
Raw Logs and Tool Outputs:


```
kali-linux-2024.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2024.4-vmware-amd64
Home X
1 2 3 4
Nmap scan report for 192.168.40.1
[*] Nmap: Host is up (0.0019s latency).
[*] Nmap: Not shown: 992 filtered tcp ports (no-response)
[*] Nmap: PORT      STATE SERVICE      VERSION
[*] Nmap: 135/tcp    open  msrpc        Microsoft Windows RPC
[*] Nmap: 139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
[*] Nmap: 445/tcp    open  microsoft-ds?
[*] Nmap: 903/tcp    open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
[*] Nmap: 5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
[*] Nmap: 8000/tcp   open  http         Splunkd httpd
[*] Nmap: 8089/tcp   open  ssl/http     Splunkd httpd
[*] Nmap: 16992/tcp  open  http         Intel Active Management Technology User Notification Service httpd 15.0.50.2633
[*] Nmap: Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:15.0.50.2633
[*] Nmap: Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 38.18 seconds
msf6 > Interrupt: use the 'exit' command to quit
msf6 > ]
[-] Unknown command: ]. Run the help command for more details.
msf6 >
msf6 > Metasploit Documentation: https://docs.metasploit.com/
[-] Unknown command: Metasploit. Run the help command for more details.
msf6 >
msf6 > msf6 > systemctl start postgresql
[-] Unknown command: msf6. Run the help command for more details.
msf6 > [*] exec: systemctl start postgresql
[-] Unknown command: [*]. Run the help command for more details.
msf6 >
msf6 > msf6 > msfdb init
[-] Unknown command: msf6. Run the help command for more details.
msf6 > [*] exec: msfdb init
[-] Unknown command: [*]. Run the help command for more details.
msf6 >
msf6 > [i] Database already started
[-] Unknown command: [i]. Run the help command for more details.
msf6 > help
```